

Gaussian Fitting with Low-order Power Functions Approximation

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Abstract—Gaussian fitting plays important role in many applications, such as localization microscopy, image scanning microscopy, which require to localize large scale of Gaussian patterns localization for reconstructing a super-resolution image. In experiment, the measured data for Gaussian fitting could be very noisy, which could affect the performance of fitting algorithm seriously. Therefore, efficient and robust Gaussian fitting is essential for these applications. In this letter, we show that a complex function like Gaussian function can be fitted by a constructed approximate function with low-order segmented functions, which could have advantage in calculation efficiency. We show that the combination of a quadratic function and its inverse function are already enough to generate accurate fitting for Gaussian model approximation. The simulation shows that the estimated position are very close to true value and the efficiency of the fitting algorithm can be improved by 6 folds.

Keywords—Gaussian fitting; function approximate; parameter initial-value estimation; efficient fitting algorithm

I. INTRODUCTION

Gaussian fitting is widely used in many applications. Take for example, photoactivated localization microscopy (PALM) [1, 2] and stochastic optical reconstruction microscopy (STORM) [3, 4] have to localize large amount of experimentally measured focal light-spots to determine the exact coordinate of each molecule to reconstruct an image with super-high resolution; image scanning microscopy (ISM) needs to localize all measured confocal light-spots to determine each scanning position for super-resolution image reconstruction [5-7]. Fluorescence Correlation Spectroscopy use Gaussian fitting for quantitative analysis of molecule property [8], accurate lattice constant deviation determination of a material [9], etc.

So far, several Gaussian fitting techniques have been available for application. The basic Gaussian fitting method is the Caruanas' algorithm [10], which transforms the Gaussian fitting problem into a least-squares fitting problem by taking logarithm on both sides of Gaussian model function. By rearranging the resulting equation, a linear function between the data set and the parameters to be solved can be obtained. Even though this algorithm has been demonstrated computationally efficient, it is demonstrated to be very sensitive to noise. To improve the performance of Caruanas' algorithm, an iterative scheme of Caruanas' algorithm is developed [11], which is much more robust to

noise. However, the drawback of these methods is that background is not taken into consideration in the fitting model. Therefore, the fitting accuracy could be affected by the background seriously, especially for very noisy data with strong background.

Another type of Gaussian fitting method is non-linear optimization based fitting technique, which establishes an accurate fitting model and then solves the fitting problem with a non-linear optimization method. Generally, this type of fitting method is much more powerful and widely used in practice for its high accuracy and straightforwardness, which however requires a good initial estimate and could results in higher cost of computing time. Even though there have been many Gaussian fitting software or packages have been developed and available on many platforms, such as Matlab fitting toolbox, for application, it is still attractive and necessary to explore new approach with high efficiency.

In this paper, a Gaussian fitting technique based on low-order power functions approximation is proposed. In section II, we model the Gaussian fitting problem based on the proposed technique and describe our algorithm. In section III, the proposed algorithm is validated by simulation. Finally, we summary our work in section IV.

II. THEORY

A. Problem Modeling

Gaussian fitting is the process of determining the parameters of a Gaussian model, so that the estimated model matches the given observed data set, which usually is corrupted by noise, with high accuracy.

For accurate Gaussian fitting problem, the model function of the 1D Gaussian function can be written as

$$f(x) = Ae^{-\frac{(x-u)^2}{2\sigma^2}} + C \quad (1)$$

where A is the amplitude, a is the coordinate of the peak, σ is the parameter regarding the width of the Gaussian function, and C denotes the background. For the sake of analysis, let us define $\theta = [A, u, \sigma, C]^T$. The fitting problem actually can be described as an optimization problem.

Basically, the distance between the estimated model with the observed data for fitting can be defined by

$$E(\theta) = \frac{1}{2} \sum_i r_i^2 = \frac{1}{2} \sum_i [f(\theta)_i - Z_i]^2 \quad (2)$$

where i is the index of pixels, $r = [r_1, r_2, \dots, r_N]^T$ denotes the residual errors for all pixels, Z denotes the observed noisy data. Minimizing the above objective function gives the solution of the parameters of fitting function, i.e.

$$\hat{\theta} = \operatorname{argmin}_{\theta} E(\theta) \quad (3)$$

Apparently, the above problem is a typical non-linear optimization problem, which can be solved by many existing non-linear optimization algorithms, such as Gauss-Newton algorithm [12], Levenberg-Marquardt (LM) method [13, 14], and Quasi-Newton method.

Differ from the traditional Gaussian fitting approach, we introduce a segmented approximate function that contains only a few low-order basic power functions to realize Gaussian fitting, as the lower order functions could make problem much easier to solve.

In general, a continuous function can be approximated by

$$y(x) = \sum_0^N a_n x^n \quad (4)$$

It would requires many terms to approximate the Gaussian function with limited terms of power function, which might result in high cost of calculation. Besides, the edge usually cannot be fitted well by general power functions.

A better way would be using a piecewise function that has similar profile as a Gaussian function. Take for example, the top of the Gaussian function can be approximated by a polynomial function with even-power, and the bottom can be approximated by another polynomial function with minor even-power function, i.e.

$$y(x) = \begin{cases} \sum_0^M -a_m x^{2m} & |x| \leq \mu \\ \sum_0^N b_n x^{-2n} & |x| > \mu \end{cases} \quad (5)$$

We here consider the simplest form,

$$y = \begin{cases} -ax^2 + \lambda & |x| \leq \mu \\ \frac{b}{x^2} & |x| > \mu \end{cases} \quad (6)$$

whose derivate can be obtained easily by

$$y' = \begin{cases} -a(x-u)^2 + \lambda & |x| \leq \mu \\ \frac{b}{(x-u)^2} & |x| > \mu \end{cases} \quad (7)$$

Since the Gaussian function is continuous function, the approximated function has to be guaranteed continuous everywhere to fit the Gaussian model as perfect as possible. In theory, it just need to let the value of the two functions and their 1st-order derivatives identical at the break point, namely,

$$\begin{cases} -ax^2 + \lambda = \frac{b}{x^2} \\ -2ax = -\frac{2b}{x^3} \end{cases} \quad (8)$$

from which, we have

$$\begin{cases} x = \sqrt[4]{\frac{b}{a}} \\ \lambda = 2\sqrt{ab} \end{cases} \quad (9)$$

which indicates that, given parameter a and b , the break point will take place at $x = \sqrt[4]{\frac{b}{a}}$ and the peak amplitude of the approximation function is $\lambda = 2\sqrt{ab}$.

For the case of 2D, the approximated function can be written by

$$f(x, y) = \begin{cases} -a[(x-u)^2 + (y-v)^2] + 2\sqrt{ab} + c, & |R| \leq \mu \\ \frac{b}{(x-u)^2 + (y-v)^2} + c, & |R| > \mu \end{cases} \quad (10)$$

where $\mu = \sqrt[4]{\frac{b}{a}}$, (u, v) denotes the coordinate of the center of the function, c denotes the background and R denotes the distance from (u, v) to (x, y) . As the form of (10) is so simple, it is very easy to obtain the derivate of the approximated function to each parameter, which can be calculated by

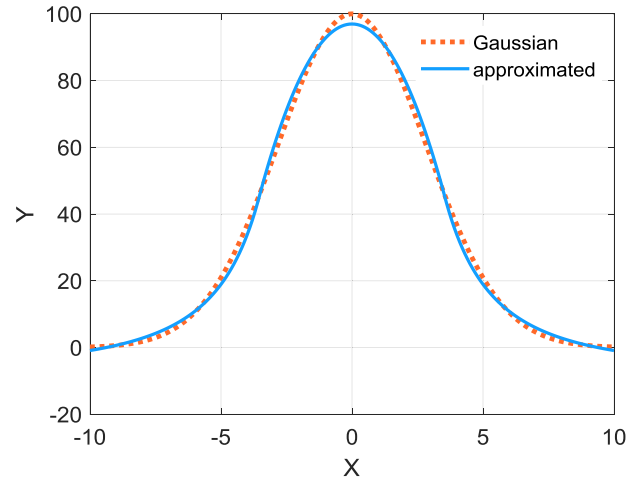


Figure 1. Standard Gaussian function fitted by formula (6).

$$f'(a) = \begin{cases} -[(x-u)^2 + (y-v)^2] + \sqrt{\frac{b}{a}}, & |R| \leq \mu \\ 0, & |R| > \mu \end{cases} \quad (11)$$

$$f'(b) = \begin{cases} \sqrt{\frac{a}{b}}, & |R| \leq \mu \\ \frac{1}{(x-u)^2 + (y-v)^2}, & |R| > \mu \end{cases} \quad (12)$$

$$f'(u) = \begin{cases} 2a(x-u), & |R| \leq \mu \\ \frac{2b(x-u)}{[(x-u)^2 + (y-v)^2]^2}, & |R| > \mu \end{cases} \quad (13)$$

$$f'(v) = \begin{cases} 2a(y-v), & |R| \leq \mu \\ \frac{2b(y-v)}{[(x-u)^2 + (y-v)^2]^2}, & |R| > \mu \end{cases} \quad (14)$$

$$f'(c) = 1 \quad (15)$$

respectively, and the gradient then can be obtained by $g = [f'(a), f'(b), f'(u), f'(v), f'(c)]^T \cdot r$.

This transformation significantly simplifies the calculation of optimization algorithm since it does not involves any exponential terms but only 2nd-order power functions. The routine of implementing the Gaussian fitting algorithm with the proposed technique is given in Algorithm 1.

Algorithm 1. Gaussian Fitting algorithm

input: data for fitting Z , threshold value τ , maximum iterations K ; set an initial value for each parameter $\theta_0 = (a_0, b_0, u_0, v_0, c_0)$;

while ($e_k > \tau$ && $k < K$)

$g_k = [f'(a), f'(b), f'(u), f'(v), f'(c)]^T \cdot r$;

Calculate the descent direction d_k by a non-linear optimization technique with gradient g_k ;

Determine the step size α by a line-searching method;

$\theta_{k+1} = \theta_k + \alpha d_k$;

$e_k = \|\theta_k - \theta_{k+1}\|_2$;

$k = k + 1$;

end

Output: θ_{k+1}

B. Solving the Involved Optimization Problem

After establishing the fitting model, the next problem is to solve the problem of minimizing the objective function. Basically, a non-linear optimization algorithm can be used to solve a minimizing problem. The basic non-linear optimization method is Newton method, which is originally design to solving the problem of zero-finding problem with high efficiency and accuracy due to its quadratic convergence property. However, there is a drawback of Newton method for solving non-linear optimization that it requires calculating the 2th order derivative of the objective function, which is very complicated and time-consuming in practice. A simple alternative of Newton method is Gauss-Newton method, which simply substitutes the second order derivative in Hessian matrix with a constant, and use only the first-order derivative to approximate Hessian matrix. Therefore, it simplifies the iteration formula and significantly reduces the algorithm complexity compared to Newton method. The problem of Gaussian-Newton method is that it is actually a local optimization method, which cannot guarantee global convergence. An enhanced Gauss-Newton method is the well-known LM method. By introducing a damping factor to control the step-size adaptively, LM algorithm works like a hybrid algorithm of Steepest Descend method [15] and Gaussian Newton method. Even though LM algorithm is capable of avoiding local stagnation problem, it is still not robust enough for noisy data and it usually requires an initial estimate very close to the true optimal solution. In comparison, the Quasi-Newton method would be a better option for solving non-linear optimization problem, which is a global optimization technique and has been proved much more efficient than the Newton method and robust to the above-mentioned basic techniques. Therefore, we here employ the most popular quasi-Newton method,

Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm [16], to implement the Gaussian fitting algorithm.

BFGS algorithm takes advantage of the previous calculated gradient information to estimate an approximate Hessian matrix. As only gradient is needed, the algorithm complexity is much lower than that of Newton method, but just similar to that of a general gradient-based non-linear optimization technique such as LM algorithm. The Global convergence of BFGS algorithm is guaranteed by introducing line searching.

The implementation of BFGS algorithm is quite simple. The iterative formula of BFGS algorithm can be written by

$$\theta_{k+1} = \theta_k + \alpha_k p_k \quad (16)$$

where p_k is the searching direction, and α_k is the step-size. Basically, the step-size can be obtained by a line searching method, such as backtracking line-search method. The searching direction is calculated by

$$p_k = -H_k \nabla g_k \quad (17)$$

where H_k is the approximated Hessian matrix, which is calculated iteratively by

$$H_{k+1} = (I - \rho_k s_k q_k^T) H_k (I - \rho_k q_k s_k^T) + \rho_k s_k s_k^T \quad (18)$$

where $s_k = \theta_{k+1} - \theta_k$, $q_k = g_{k+1} - g_k$, $\rho_k = \frac{1}{q_k^T s_k}$. The above iterative formulas of BFGS algorithm show that there are only vector and basic matrix calculation, even matrix inversion is avoided. It is indeed a big advantage in high performance optimization calculation.

C. Parameter Initial-Value Estimation

In practice, a data fitter usually requires manually inputting an initial for each parameter. The quality of estimation has significant impact to the performance of the fitting algorithm. For example, a far-away initial point to the true optimal solution could cause failure of fitting, especially for noisy data, while an appropriate estimation would result in much faster convergence of the fitting algorithm. Therefore, a high performance initial value estimation algorithm is important for robust and efficient fitting.

In order to reduce the impact of noise for initial estimation, the noisy data for fitting firstly is filtered with a low-pass filter, such as median filter, mean filter or wavelet filter. For Gaussian fitting problem, the image can be resized by bilinear interpolation, which improves the accuracy of the following parameter estimation.

The background is very common in application, take optical imaging for example. In some case, the amplitude of background could be large than the highest amplitude of the signal, therefore, the background cannot be ignored in Gaussian fitting. The easiest way would be using the average of several pixels with minimal intensity as the estimation for background.

Estimating the position of the Gaussian pattern is very simple from a background-removed data, since the data becomes smooth after low-pass filtering. The maximum pixel of the background-removed can be used for estimating the amplitude and the position of the Gaussian pattern, namely, its intensity can be selected as the initial value for A , and its coordinate for the initial value of u and v .

The peak-value λ then can be estimated by taking the pixel value at position (u_0, v_0) from the background-removed data. To avoid the impact of noise, taking the average of 2 maximum pixels would be a good option.

The parameter a can be estimated efficiently by fitting the neighbor pixels round (u_0, v_0) to model $z = -aR^2$, with the LSQ technique, which can be represented by

$$a_0 = \frac{\sum_i (z_i \cdot r_i^2)}{\sum_i r_i^4} \quad (19)$$

where $R = \sqrt{(x - u_0)^2 + (y - v_0)^2}$. Similarly, b can be estimated by fitting the rest of pixels to the model $z = \frac{b}{R^2}$ by

$$b_0 = \frac{\sum_i (z_i \cdot R_i^{-2})}{\sum_i R_i^{-4}} \quad (20)$$

To discriminate the support for the two segmented functions, the gradient along x direction can be calculated, then a peak will take place at the position of steepest ascending of a Gaussian function, which indicates the boundary of the support between segmented functions.

III. SIMULATION

We test the performance of our algorithm by simulation. The testing platform is Matlab 2018b and the computer configuration is: i7-6700HQ CPU (2.6GHz) with 16G RAM. We implement the algorithm based on Algorithm 1. The fitting problem is described as formula (2) in the Matlab code; the gradient of the objective function is calculated by the formula mentioned in Algorithm 1. In our fitting algorithm, BFGS algorithm described above is employed to solve the minimization problem. The centre coordinate (a, b) is generated with random value, while sigma and the background are set as to $\sigma = 2$, $C = 100$; Poisson noise is applied to the randomly generated data.

We vary the amplitude of the Gaussian model A and calculate the localization errors with different methods for comparison; the localization error is defined by the distance between the calculated coordinate (a, b) and the ground-truth coordinate, i.e.

$$\Delta R = \sqrt{\Delta x^2 + \Delta y^2} \quad (21)$$

The simulated noisy images with different amplitudes are just as Figure 2 shows.

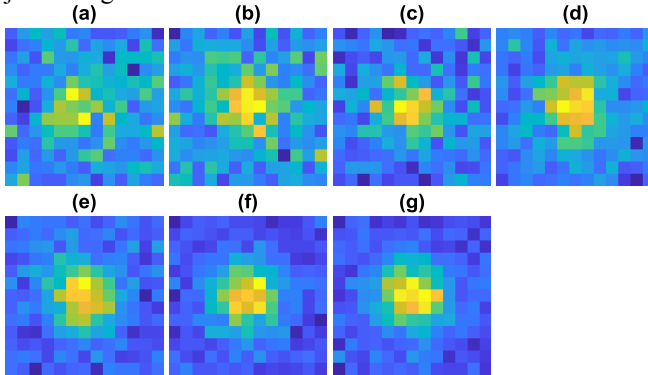


Figure 2. A sample of the noisy images with different amplitudes with background 100. (a) $A=30$, (b) $A=40$, (c) $A=50$, (d) $A=70$, (e) $A=90$, (f) $A=120$, (g) $A=150$.

The result of our simulation is just as Fig.3 shows. It shows that the fitting accuracy of the proposed technique is quite close to the standard Matlab fitter, while the error by the Iterative LSQ algorithm [17] is about 3 times larger. The reason is that Iterative LSQ algorithm assumes that the Gaussian model does not contain background, therefore it fits the data using a standard Gaussian function without constant term. Actually, the background is quite strong in our testing case, which actually is quite common in experiment. The result also proves that the Gaussian approximation with lower-order functions is quite well, which does not affect the fitting accuracy so much, while shows significant efficiency improvement. The CPU time for the calculation is just as table 1 shows.

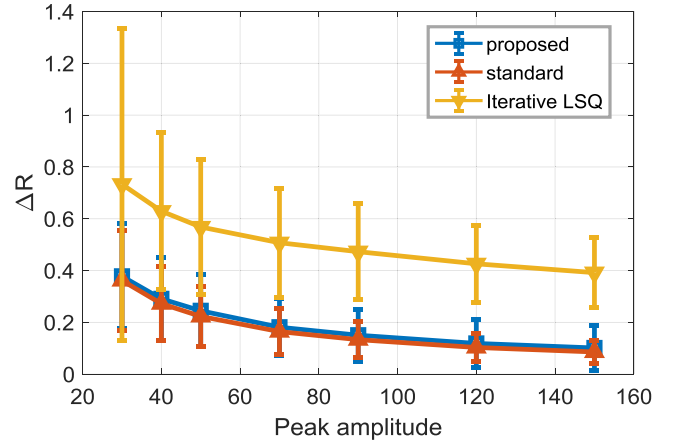


Figure 3. Fitting errors ΔR vs A with different methods.

TABLE I. CPU TIME FOR CONVERGENCE

Method	Proposed	Standard	Iterative LSQ
Time	5.0ms	30ms	0.55ms

It shows that Iterative LSQ algorithm is the fastest algorithm, which takes less 1ms for a single fitting; even though the proposed technique is slower, its efficiency is still much higher than the standard Matlab fitter, which shows about 6 folders efficiency improvement. Apparently, the efficiency improvement actually benefits from low-order power function approximation; as the approximated function is much less complex than an exponential function, the fitting problem becomes easier to solve.

IV. CONCLUSION

In this paper, we propose an efficient accurate Gaussian fitting technique. A Gaussian function can be approximated by several low-order power functions, by which the complex Gaussian fitting problem is simplified. We also propose a simple parameter initial value estimation technique to improve the performance of the proposed algorithm. Simulation shows that the accuracy of the proposed technique is quite closed to that of the standard method,

while the efficiency is much higher, which shows its potentiality for application.

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